Beyond real solutions

$$x^2 = -9$$

Why does this equation have no real solution?

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The imaginary unit

The imaginary unit i is the principal square root of -1; its square is -1.

$$i = \sqrt{-1}$$

Why is i not a real number?

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A complex number

A complex number has the form a + bi, where a and b are real.

$a + bi$

How are real numbers included among complex numbers?

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Identify the parts

$-4 + 7i$

What are this number’s real part and imaginary part?

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Square roots of negative numbers

For a positive real number a, move the negative sign out of the square root as i. Then simplify the positive radical.

$$\sqrt{-a} = i\sqrt{a} \qquad (a > 0)$$

Which simplification skills from Section 1.3 still apply?

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A negative perfect square

$$\sqrt{-36}$$

How can we write the principal square root as a multiple of i?

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A negative radicand

Use the principal complex square root.

$$\sqrt{-50}$$

How can we write this radical in standard complex form?

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A negative radicand with a square factor

$$\sqrt{-12}$$

How can we write the principal square root as i times a simplified radical?

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Two square roots

$$z^2 = -25$$

What are both complex solutions?

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One principal root or two solutions?

$\sqrt{-12}$ and $x^2 = -12$

Simplify the radical, then solve the equation. Which answer needs a plus-or-minus sign?

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